Algorithm Course Outline

Course Title: **SWE 221: Algorithm**

Prerequisites: Basic Programming Knowledge, Data Structures

Module 1: Introduction to Algorithms

1. Definition and characteristics of algorithms.
2. Algorithmic problem-solving process.
3. Basics of pseudocode and flowcharts.
4. Types of algorithms:
    a. Divide and Conquer
    b. Greedy Algorithms
    c. Dynamic Programming
    d. Backtracking
    e. Brute Force

Module 2: Analysis of Algorithms

tions.

2. Space Complexity: Trade-offs between time and space.
3. Asymptotic analysis.

Module 3: Sorting and Searching Algorithms

1. **Sorting**:
    a. Bubble Sort, Selection Sort, Insertion Sort (Elementary Sorting).
    b. Merge Sort, QuickSort, Heap Sort (Efficient Sorting).
    c. Counting Sort (Non-comparison-based Sorting).
2. **Searching**:
    a. Linear Search.
    b. Binary Search.

Module 4: Graph Algorithms

1. Basics of graphs: Representations (Adjacency Matrix/List).
2. Breadth-First Search (BFS) and Depth-First Search (DFS).

3. Shortest path algorithms:
    a. Dijkstra's Algorithm.
    b. Bellman-Ford Algorithm.
    c. Floyd-Warshall Algorithm.
4. Minimum Spanning Tree:
    a. Kruskal's Algorithm.
    b. Prim's Algorithm.

Module 5: Greedy Algorithms

1. Characteristics of greedy algorithms.
2. Activity Selection Problem.
3. Fractional Knapsack Problem.
4. Huffman Encoding.

Module 6: Dynamic Programming

1. Principles of dynamic programming.
2. Fibonacci Sequence.
3. Longest Common Subsequence (LCS).

4. 0/1 Knapsack Problem.

5. Matrix Chain Multiplication.

6. Longest Increasing Subsequence (LIS).

Module 7: Backtracking and Recursion

1. Principles of backtracking.

2. N-Queens Problem.

Reference Books and Resources

1. Introduction to Algorithms by Thomas H. Cormen, Charles E., Leiserson, Ronald L. Rivest, Clifford Stein.

2. Data Structures and Algorithms Made Easy by NarasimhaKarumanchi.

3. Algorithms by Robert Sedgewick, Kevin Wayne

4. Online platforms like GeeksforGeeks, LeetCode, and HackerRank for practice.